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Lightweight Real-Time Intrusion Detection for IoT Networks Using TSANet Deployed on Raspberry Pi with Live Traffic Monitoring

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Abstract

Intrusion Detection Systems (IDS) play a crucial role in keeping IoT networks secure. However, many deep learning–based IDS solutions require too much computing power to run on devices with limited resources, like those found at the edge of networks. In this paper, we introduce TSANet—a lightweight Temporal–Spatial Attention Network designed for real-time intrusion detection on low-power devices, such as the Raspberry Pi 5. TSANet combines a 1D Convolutional Neural Network to learn spatial features, an LSTM to understand temporal patterns in data, and attention mechanisms to focus on the most important parts of network traffic. We trained and tested TSANet on the CICIDS2017 dataset, then applied pruning and quantization-aware training to make the model smaller and faster. Our results show that the optimized model achieves 98.3% accuracy in offline tests and 96.8% accuracy in real-time detection on the Raspberry Pi, with an inference delay of just 8–12 milliseconds per data flow. These findings demonstrate that TSANet is practical and effective for protecting IoT devices at the edge.

Keywords: Intrusion Detection System; IoT Security; Edge Computing; Raspberry Pi; Deep Learning

1 Introduction

This research is in line with the United Nations' Sustainable Development Goal 9 (SDG 9), which focuses on the development of resilient infrastructure, inclusive and sustainable industrialization, and innovation-driven by secure and reliable digital platforms. The increasing popularity of Internet of Things (IoT) technology has resulted in the widespread adoption of IoT devices in smart homes, healthcare, industrial automation, and smart cities. These devices are continuously exchanging data through networks and are often deployed in open or untrusted environments, making them prime targets for cyber-attacks such as denial-of-service (DoS), distributed denial-of-service (DDoS), malware injection, and data breaches. The need to secure these IoT networks has become a pressing concern.

Intrusion Detection Systems (IDS) are commonly employed to monitor network traffic and detect malicious activity. Traditional IDS systems employ signature-based or rule-based systems, which are efficient in detecting known attacks but are ineffective against unknown or evolving threats. In recent years, deep learning-based IDS systems have shown excellent detection capabilities by learning complex patterns from network traffic. However, most existing deep learning-based models are computationally intensive and are designed to be executed on powerful servers or cloud platforms.

Deploying such models directly on IoT edge devices is difficult due to limited processing power, memory, and energy availability. Transmitting raw network data to centralized servers also introduces latency and raises privacy concerns. These limitations highlight the need for lightweight, real-time intrusion detection systems that can operate directly on edge devices without relying on external computation.

To address these challenges, this work proposes TSANet, a lightweight Temporal–Spatial Attention Network designed for efficient deployment on resource-constrained hardware. The proposed system performs complete intrusion detection at the edge, including traffic analysis and real-time inference, and is validated through deployment on a Raspberry Pi 5 using both benchmark datasets and live network traffic. The results demonstrate that effective intrusion detection can be achieved at the IoT edge with low latency and high accuracy.

2 Related Work

Internet of Things networks are getting a lot of attention from researchers when it comes to detecting intrusions. This is because many devices are being connected to the internet in smart cities, smart homes and industrial areas.When we look at the research that has been done far we can see that it falls into a few main categories. These are the ways of detecting intrusions methods that use deep learning to power Intrusion Detection Systems and solutions that are lightweight enough to be used on the edge of Internet of Things networks. You can see a summary of this research, in Table 1.

2.1 Traditional and Early IDS Approaches

Early intrusion detection research primarily relied on conventional machine learning and rule-based techniques to identify malicious activities in network traffic. These approaches established the foundation for IDS development but struggle to scale effectively in heterogeneous and dynamic IoT environments. The importance of secure digital infrastructure has also been emphasized globally through initiatives such as the United Nations Sustainable Development Goal 9, which highlights resilient and secure infrastructure as a key requirement for sustainable development [1].

2.2 Deep Learning-Based Intrusion Detection Systems

To address the limitations of traditional methods, deep learning techniques have been widely adopted for intrusion detection in IoT networks. Idrissi et al. proposed a deep learning–based host IDS optimized for IoT environments, achieving improved detection accuracy through architectural optimization [2]. Similarly, Saba et al. introduced an anomaly-based IDS using deep learning models to enhance attack detection performance [3].

Hybrid architectures combining convolutional and recurrent models have also been explored. Gueriani et al. integrated CNN and LSTM networks to capture both spatial and temporal traffic characteristics [4], while Otoum and Nayak proposed a deep learning IDS framework emphasizing sequential learning for IoT security [5]. Survey studies further highlight the effectiveness of deep learning for IoT IDS while noting challenges related to model complexity and deployment feasibility [6, 7].

2.3 Lightweight and Edge-Based IDS Solutions

Recent research increasingly focuses on deploying intrusion detection systems directly on edge devices to reduce latency, bandwidth usage, and privacy risks. Kayode et al. proposed a Temporal–Spatial Attention Network (TSANet) for DoS attack detection, demonstrating the benefit of attention mechanisms in highlighting critical traffic features [8]. Racherla et al. developed a real-time deep IDS for IoT nodes, showing the feasibility of deep learning deployment in constrained environments [9].

Several studies specifically investigate Raspberry Pi–based IDS deployments. Dahiya et al. demonstrated a Raspberry Pi–based IDS for IoT networks [10], while Olakanmi and Olatunji proposed an optimized ensemble deep learning IDS achieving improved accuracy with reduced resource usage [11]. Adrian et al. designed an adaptive IDS for smart home security on Raspberry Pi platforms [12], and Gupta et al. explored deep learning–based IDS in 5G-enabled IoT networks, emphasizing low-latency detection requirements [13]. Despite these advances, many approaches lack fully integrated real-time traffic capture and inference on low-power hardware.

Table 1: Summary of IoT Intrusion Detection Systems

Author	**Methodology**	**Dataset**	**Feature Extraction**	**Results**
Idrissi et al. (2022) [2]	Optimized deep learning HIDS for IoT edge	IoT system logs	Host-level statistical features	Improved accuracy with reduced overhead on edge devices
Saba et al. (2022) [3]	Anomaly-based deep learning IDS	IoT traffic	Statistical flow features	High detection rate for IoT attacks
Gueriani et al. (2023) [4]	Hybrid CNN–LSTM IDS	Simulated IoT traffic	Spatial and temporal flow features	Enhanced detection accuracy over single models
Otoum and Nayak (2022) [5]	Deep learning–based IDS (DL-IDS)	IoT network traffic	Sequential traffic features	Effective detection of complex attacks
Kayode et al. (2025) [8]	Temporal–Spatial Attention Network (TSANet)	Network traffic traces	Attention-weighted flow features	High DoS detection accuracy using attention mechanisms
Racherla et al. (2024) [9]	Real-time deep IDS for IoT	IoT network traffic	Flow-based statistical features	Low-latency detection on IoT nodes
Dahiya et al. (2024) [10]	Raspberry Pi–based IDS	IoT traffic	Basic flow statistics	Feasible edge deployment with limited evaluation
Olakanmi and Olatunji (2024) [11]	Optimized ensemble deep learning IDS	IoT traffic	Reduced feature set	Improved accuracy with lower resource usage
Adrian et al. (2025) [12]	Adaptive ML-based IDS	Smart home traffic	Network flow features	Adaptive detection for smart home security
Gupta et al. (2022) [13]	Deep learning IDS for 5G-IoT	IoT over 5G networks	Traffic flow features	Low-latency detection in 5G-enabled IoT

Author	Methodology	Dataset	Feature Extraction	Results
Tsimenidis et al. (2022) [6]	Survey on DL-based IoT IDS	Multiple benchmarks	Comparative feature analysis	Identified challenges in DL IDS deployment
Morshedi et al. (2024) [7]	Deep learning IDS for IoT security	IoT datasets	Flow-level features	High accuracy but limited edge evaluation
Shone et al. (2018) [14]	Deep autoencoder-based IDS	NSL-KDD	Learned latent representations	Improved detection of unseen attacks

2.4 Research Gap

Although many intrusion detection systems have been proposed for IoT networks, most of the existing solutions are based on computationally complex deep learning models that are designed for cloud or server environments and are not suitable for real-time execution on edge devices. Many studies have shown high accuracy in intrusion detection only in offline settings and have not tested the latency, resource consumption, and robustness of the system in real-world settings. Moreover, many solutions have been proposed for either spatial or temporal feature learning, but not both. Although model compression and federated learning have been explored.

3 Proposed Methodology

This section introduces the proposed lightweight intrusion detection system design based on the Temporal-Spatial Attention Network (TSANet). The main contribution of this research work is the real-time implementation of an attention-based deep learning IDS system running entirely on a Raspberry Pi edge device based on live network traffic. Unlike traditional IDS system pipelines that use centralized processing, the proposed IDS system design captures traffic, extracts features, makes predictions, and generates alerts entirely at the edge

3.1 System Overview

The entire process of the proposed system is shown in Figure 3. The network packets are captured in real-time from the live network, transformed into bidirectional flows, processed into numerical features, and classified by the trained TSANet model on the Raspberry Pi device based on the confidence level of the prediction.

3.2 Data Collection and Feature Engineering

The data for network traffic is collected in two stages. In the offline training stage, the CICIDS2017 dataset is employed owing to its realistic network traffic and multiple classes of attacks. In the real-time testing stage, the network traffic is directly captured on the Raspberry Pi, which is set up as a Wi-Fi access point.

Each network flow is described by 32 features selected from bidirectional flows, such as packet-level information, flow duration, byte counts, and inter-arrival times. These features are selected to achieve a balance between detection accuracy and computational complexity, making them suitable for edge devices. The features are scaled using Min-Max normalization to ensure numerical stability during inference.

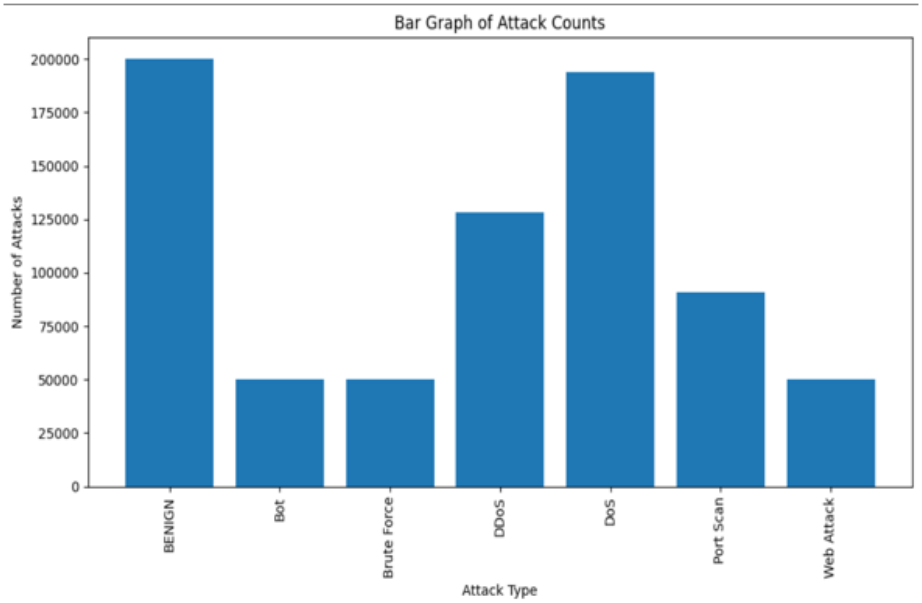

Figure 1: Distribution of benign and attack samples in the CICIDS-2017 dataset

Figure 1 shows the distribution of benign and attack samples in the CICIDS-2017 dataset. This figure is used as the starting point for the proposed TSANet-based intrusion detection system. The normalized flow features extracted from the CICIDS-2017 dataset and real-time traffic (as mentioned in the previous subsection) are used as input to the TSANet model for learning temporal features and attack classification.

3.3 TSANet Architecture

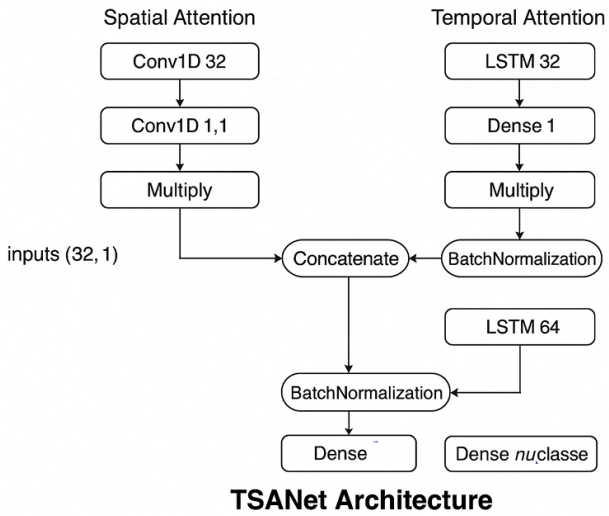

Figure 2: TSANet architecture illustrating spatial and temporal attention mechanisms

The heart of the proposed system is the TSANet model, whose architecture is depicted in Figure 2. TSANet leverages convolutional, recurrent, and attention mechanisms to adequately capture the spatial and temporal patterns in network traffic.

The spatial attention mechanism consists of a one-dimensional convolutional layer with 32 filters to extract local feature interactions from the input feature vector. A subsequent 1×1 convolution is used to compute attention weights, which are applied via element-wise multiplication to highlight discriminative features and attenuate non-informative ones.

Concurrently, the temporal attention mechanism uses an LSTM layer with 32 hidden units to extract sequential patterns over time. A dense layer with a sigmoid activation function is used to compute temporal attention weights, enabling the network to selectively attend to important time instances in each flow. Batch normalization is used to enhance training stability and convergence.

The outputs of the two attention mechanisms are concatenated and fed into an LSTM layer with 64 units to learn spatio-temporal representations. Finally, fully connected layers project the learned features into the output space, yielding class probabilities for benign and attack traffic.

3.4 Model Training and Optimization

TSANet is trained using the supervised learning approach with categorical cross-entropy loss as the loss function. Taking the input feature sequence

$X = \{x_1, x_2, \ldots, x_T\}$ and the corresponding ground-truth label y, the optimization goal is formulated in Eq. (1).

$$\mathcal{L} = -\sum_{c=1}^{C} y_c \log(\hat{y}_c) \tag{1}$$

where $\hat{y}_c$ represents the predicted probability for class c.

To facilitate efficient edge device deployment, the trained model is pruned and quantization-aware training is performed to optimize the model size and inference speed without compromising model accuracy. The optimized model is then exported with TorchScript for

3.5 Edge Deployment on Raspberry Pi

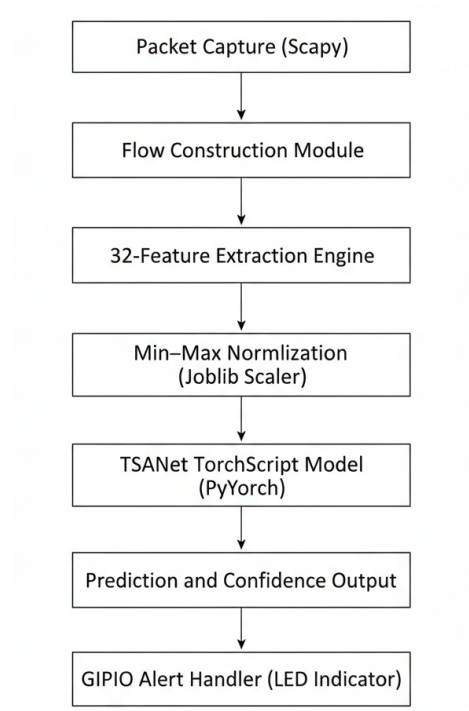

Figure 3: End-to-end system workflow for real-time traffic capture, feature extraction, and edge inference

The optimized TSANet model is then deployed on a Raspberry Pi 5, designed to function as a live network gateway. Packet capture is done using Scapy, followed by flow reconstruction and feature extraction on the same device. Inference is done locally using the TorchScript model.

Figure 3 illustrates the complete runtime pipeline, including packet capture, feature processing, model inference, and alert generation. Upon detecting malicious activity, the system outputs prediction confidence and triggers a GPIO-based LED indicator for immediate physical alerting. This fully autonomous design eliminates reliance on external servers and preserves data privacy.

Overall, the proposed methodology demonstrates that effective intrusion detection can be achieved directly at the IoT edge by integrating attention-driven deep learning with lightweight deployment strategies.

4 Results and Discussion

This section discusses the experimental results of the proposed TSANet-based intrusion detection system on benchmark datasets and real-time deployment on a Raspberry Pi edge device.

4.1 Offline Evaluation on CICIDS2017

The proposed TSANet model was first evaluated offline using the CICIDS2017 dataset. Performance was measured using standard classification metrics including accuracy, precision, recall, and F1-score.

Table 2 compares the proposed TSANet model with commonly used baseline models. The proposed model demonstrates the highest overall accuracy of 98.3%, outperforming traditional machine learning and deep learning models. This is because the proposed model integrates the spatial and temporal attention mechanism, allowing the model to focus on the most important features and time steps.

Table 2: Comparison of TSANet with Baseline Models

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**
Random Forest	94.2	93.7	92.5	92.8
CNN	95.6	95.1	94.8	94.9
LSTM	96.2	96.0	95.7	95.8
CNN–LSTM	97.0	96.9	96.4	96.6
Proposed TSANet	**98.3**	**98.6**	**97.9**	**98.1**

4.2 Effect of Model Pruning

To optimize TSANet for edge deployment, structured pruning was applied. Figure 4 shows the relationship between pruning rate and classification accuracy. The results indicate that accuracy remains stable up to moderate pruning levels, demonstrating that TSANet retains robustness even after model compression.

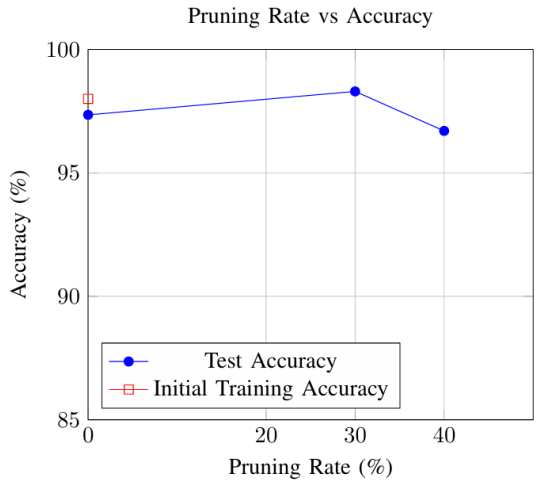

Figure 4: Impact of pruning rate on classification accuracy

4.3 Analysis of Confusion Matrix

The confusion matrix of the optimized TSANet model on the CICIDS2017 dataset is shown in Table 3. It can be seen that the detection rates are very high for volumetric attacks such as DoS and DDoS. There is slight confusion between classes for sparse classes such as Heartbleed, which is mostly due to a lack of evidence at

Table 3: Confusion Matrix of Optimized TSANet on CICIDS2017 (%)

Actual / Predicted	Benign	Bot	BF	DDoS	DoS	Heartbleed	Infiltration
Benign	98.1	0.2	0.3	0.1	0.8	0.0	0.5
Bot	0.4	97.6	1.2	0.0	0.3	0.0	0.5
Brute Force	0.6	0.8	96.4	0.0	1.7	0.0	0.5
DDoS	0.0	0.0	0.0	98.9	1.1	0.0	0.0
DoS	0.4	0.1	0.3	1.5	97.3	0.1	0.3
Heartbleed	0.0	0.0	0.0	0.0	2.5	95.8	1.7
Infiltration	1.3	0.4	0.0	0.0	0.7	0.2	97.4

4.4 Real-Time Deployment and Live Traffic Analysis

To address the shortcomings of the Mininet-based experiments, the optimized TSANet model was deployed on a Raspberry Pi 5 setup as a standalone wireless access point. Laptops and mobile devices were connected to the access point, and real Wi-Fi traffic was generated through web browsing, messaging, YouTube

videos, system updates, and background processes. As all packets were routed through the Raspberry Pi, complete bidirectional flows were captured, allowing for accurate feature extraction and real-time prediction.

During real-time operation, each network flow was denoted by its source IP address and port communicating with a destination IP address and port. For each such IP-to-IP flow seen at the access point, the system extracted flow-level features and fed them to the trained TSANet model, which predicted the flow as either benign or malicious along with a confidence score.

A sample snippet of the real-time alerts produced during live traffic analysis is provided below. The output illustrates how traffic from one IP address to another is analyzed and marked by the system in real time.

```
===== ALERT =====
Flow: ('163.70.144.8', 443) → ('192.168.50.100', 45190)
Prediction: BENIGN (conf=0.5827)
=================

===== ALERT =====
Flow: ('192.168.50.140', 8080) → ('192.168.50.33', 41076)
Prediction: Heartbleed (conf=0.7354)
=================

===== ALERT =====
Flow: ('192.168.50.140', 52550) → ('34.107.221.82', 80)
Prediction: BENIGN (conf=0.5482)
================-

===== ALERT =====
Flow: ('192.168.50.140', 8080) → ('192.168.50.33', 41076)
Prediction: DDoS (conf=0.7714)
=================
```

In the above output, each alert is related to a single network flow captured at the access point, where the flow is sent from the source IP address to the destination IP address. The system predicts whether the captured flow is benign or belongs to a certain type of attack, along with a confidence level that represents the certainty of the prediction. Benign flows due to normal user traffic are correctly classified, while attack flows produce high-confidence alerts.

5 Conclusion and Future Work

In this paper, we proposed TSANet, a lightweight Temporal-Spatial Attention Network for real-time intrusion detection in IoT networks, which can be efficiently executed on edge devices. Through the combination of spatial and temporal attention mechanisms with CNN and LSTM layers, our proposed system is capable of extracting distinctive traffic patterns with low computational

complexity. Experimental results on the CICIDS2017 dataset showed that our proposed system outperformed the baseline models with a high offline accuracy of 98.3%. Real-time experiments on a Raspberry Pi setup as a Wi-Fi access point confirmed the feasibility of our proposed system, which exhibited low inference latency of 8-12 ms per flow and stable resource consumption during real-time traffic analysis. Our proposed system was able to detect both normal and malicious behaviors without the need for cloud support. Future work includes the extension of our proposed system to handle more attack types and adaptive learning for evolving threats, as well as experiments on large-scale distributed IoT networks.

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